

A Comparative Analysis of Classical Machine Learning Methods and Variational Quantum Methods for Breast Tumor Classification

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 - Comparison
 - Conclusion

Our goals:

- Tumor Classification using Classical ML methods

- Tumor Classification using Variational Quantum methods

- Comparison of the results

Classical Machine Learning Basics

Two types of ML:

- Supervised (Labeled Datasets)

- Unsupervised (Hidden patterns, no labels)

Unsupervised ML:

- Clustering

- Anomaly Detection

- Dimensionality Reduction

Supervised ML:

- Regression

- Classification

We will focus on **Classification**

Binary Classification with Classical ML Methods

Binary Classification: Choose between **two categories**

Classical ML methods:

- Logistic Regression**

- Gaussian Naive Bayes (GNB)**

- Linear Discriminant Analysis (LDA)**

- Support Vector Machines (SVM)**

- Random Forests**

- Light Gradient Boosting Machine (LightGBM)**

Quantum Mechanics and Quantum Computers

Quantum Theory or Quantum Mechanics:

Mathematical **Framework** for the results of measurement of **quantum systems**

Quantum Systems:

Microscopic entities where **Newtonian** mechanics **fail** (e.g hydrogen atom)

Quantum Computer:

Computational device that **exploits quantum mechanical effects** (superposition, entanglement) to carry out **operations on information**

Uses **Qubits** instead of bits

Quantum characteristics of particles can be **utilized**

Quantum processes can **manipulate** this information

The Qubit

Fundamental building blocks of quantum computers

The quantum mechanical analogue of a classical bit

Created by manipulating and measuring systems that exhibit quantum mechanical behavior

Superconducting qubits: **superconducting** materials at **cryogenic** temperatures, offer **rapid** gate speeds and **precise** control

Trapped ion qubits: individual **ions confined** by electromagnetic fields, **long coherence** times and **high-fidelity measurements**, slower than **superconducting**

Quantum dots: tiny **semiconductors** capture a **single electron**, potential for **scalability**

Photons: particles of **light**, encode quantum information and transmit via **optical fibers**, applications in quantum **communication** and **cryptography**

The Qubit, more formally

A two-state quantum system defined by the **basis vectors** $|0\rangle$ and $|1\rangle$

A qubit may exist in a **superposition**, a mathematical **linear combination** of both basis states, $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$

$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, α, β probability amplitudes with complex values

Hermitian conjugate: $\langle\psi| = (\alpha^* \quad \beta^*)$

$|\cdot\rangle$ is called "**ket**", $\langle\cdot|$ is called "**bra**"

Upon **measurement**, Qubit **collapses** with $P(|0\rangle) = |\alpha|^2$ and $P(|1\rangle) = |\beta|^2$

Thus, it must hold $|\alpha|^2 + |\beta|^2 = 1$ or $\|\psi\| = \sqrt{\langle\psi|\psi\rangle} = 1$

If not, then we do $|\tilde{\psi}\rangle = \frac{|\psi\rangle}{\|\psi\|}$

A qubit is defined as a **vector** residing in a **two-dimensional complex Hilbert** space $\mathbb{C}^2$

Bloch sphere

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right) |0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right) |1\rangle$$

$|\psi\rangle = |0\rangle$ occurs at $\theta = 0$

$|\psi\rangle = |1\rangle$ occurs at $\theta = \pi$

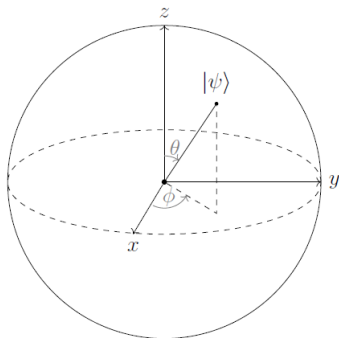


Figure: A qubit can be mapped to a point on the surface of this unit-radius sphere ($r = 1$), uniquely determined by the angular coordinates θ and ϕ

We transition qubits through **different states** with the use of **operators**

$$\hat{A}|\psi\rangle = |\phi\rangle, \hat{A}\langle\psi| = \langle\phi|$$

Vital set of operators called **Pauli** operators:

$$\hat{I}|\psi\rangle = |\psi\rangle$$

$$\hat{X}|0\rangle = |1\rangle, \hat{X}|1\rangle = |0\rangle \text{ (NOT operator)}$$

$$\hat{Y}|0\rangle = -i|1\rangle, \hat{Y}|1\rangle = i|0\rangle$$

$$\hat{Z}|0\rangle = |0\rangle, \hat{Z}|1\rangle = -|1\rangle$$

We can also express an operator as an outer product

$$(|\psi\rangle\langle\phi|)|\chi\rangle = |\psi\rangle\langle\phi|\chi\rangle = \langle\phi|\chi\rangle|\psi\rangle$$

Operators as Matrices

Another method for representing an operator is through matrix notation

For an n -**dimensional** vector space $\rightarrow n \times n$ **matrices**. We focus on $\mathbb{C}^2$, so 2×2 **matrices**

$$\hat{A} = \begin{pmatrix} \langle 0 | \hat{A} | 0 \rangle & \langle 0 | \hat{A} | 1 \rangle \\ \langle 1 | \hat{A} | 0 \rangle & \langle 1 | \hat{A} | 1 \rangle \end{pmatrix} \quad (1)$$

Pauli Operators as Matrices

Following Equation 1 we get that:

$$\hat{I} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (2)$$

$$\hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (3)$$

$$\hat{Y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (4)$$

$$\hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (5)$$

Hermitian & Unitary Operators

A given operator $\hat{A}$ is classified as **Hermitian** iff:

$$\hat{A} = \hat{A}^\dagger \quad (6)$$

$\hat{A}^\dagger$ denotes the Hermitian adjoint $\rightarrow$ transposing the matrix and taking the complex conjugate of every entry

Hermitians possess **real eigenvalues** and their **main diagonal** consists solely of **real** numbers.

Therefore they represent **physical observables**

A given operator $\hat{U}$ is classified as **Unitary** iff:

$$\hat{U}\hat{U}^\dagger = \hat{U}^\dagger\hat{U} = \hat{I} \quad (7)$$

They characterize the **temporal evolution** of a quantum state

The **Pauli** operators are **both** the Hermitian and Unitary

To explore more sophisticated and advantageous quantum algorithms, we need systems of **multiple interacting qubits**

Construction of a **composite** Hilbert space

The mathematical operation used is the Kronecker or tensor product ($\otimes$)

If H_1 and H_2 two Hilbert spaces with dimensions N_1 and N_2 , then we can form a composite Hilbert space H with dimension $N_1 N_2$, defined as:

$$H = H_1 \otimes H_2 \quad (8)$$

States derived from tensor product

The tensor product can be employed to construct a state residing within the composite space H :

$$|\psi\rangle = |\phi\rangle \otimes |\chi\rangle \quad (9)$$

For two-dimensional subspaces, the following procedure is employed to calculate the product:

$$|\psi\rangle = |\phi\rangle \otimes |\chi\rangle = \begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix} \quad (10)$$

Usually, the $\otimes$ symbol is omitted; for example, $|\phi\rangle \otimes |\chi\rangle$ may be written as $|\phi\rangle |\chi\rangle$, or even more compactly as $|\phi\chi\rangle$.

Basis vectors and operators of a composite system

If $|u_i\rangle$ and $|v_i\rangle$ are the basis vectors of H_1 and H_2 , the resulting basis for the joint space H is:

$$|w_i\rangle = |u_i\rangle \otimes |v_i\rangle \quad (11)$$

We can define operators that act upon our composite system. If $|\phi\rangle \in H_1$ and $|\chi\rangle \in H_2$, and A acting on $|\phi\rangle$ and B acting on $|\chi\rangle$. A new operator, which acts on a state $|\psi\rangle \in H$ is:

$$(A \otimes B) |\psi\rangle = (A \otimes B)(|\phi\rangle \otimes |\chi\rangle) = (A|\phi\rangle) \otimes (B|\chi\rangle) \quad (12)$$

The matrix representation is computed as:

$$A \otimes B = \begin{pmatrix} \alpha_{11}B & \alpha_{12}B \\ \alpha_{21}B & \alpha_{22}B \end{pmatrix} = \begin{pmatrix} \alpha_{11}\beta_{11} & \alpha_{11}\beta_{12} & \alpha_{12}\beta_{11} & \alpha_{12}\beta_{12} \\ \alpha_{11}\beta_{21} & \alpha_{11}\beta_{22} & \alpha_{12}\beta_{21} & \alpha_{12}\beta_{22} \\ \alpha_{21}\beta_{11} & \alpha_{21}\beta_{12} & \alpha_{22}\beta_{11} & \alpha_{22}\beta_{12} \\ \alpha_{21}\beta_{21} & \alpha_{21}\beta_{22} & \alpha_{22}\beta_{21} & \alpha_{22}\beta_{22} \end{pmatrix} \quad (13)$$

Quantum Entanglement

A composite system **cannot always** be expressed as a tensor product of individual states, only if the qubits are prepared separately and kept in isolation. If the qubits are permitted to **interact**, they form a single integrated closed system and are described as **entangled**:

Quantum Entanglement is a fundamental phenomenon and a vital resource for quantum computation

Particles sharing a common origin remain fundamentally **linked** regardless of the distance between them

Any measurement or interaction of one particle instantaneously **influences** the state of its entangled **partners**

Information can be transmitted **faster** than the speed of **light**

Quantum Gates

Transformations applied to qubit states are referred to as quantum gates, and a sequence of such operations forms a **quantum circuit**:

Quantum gates represent unitary **operations** performed on **qubits**

The terms "gate" and "unitary operator" are used **interchangeably**

A gate processing n **inputs** and outputs is represented by a $2^n \times 2^n$ **matrix**

A **single-qubit** gate is depicted as a 2×2 **matrix**

A quantum gate acting on **two qubits** requires a 4×4 **matrix** representation

The most fundamental single-qubit operation is the quantum *NOT* gate, which is equivalent to the Pauli X operator. This gate effectively swaps the basis states. Represented as a matrix in the computational basis, we have:

$$\hat{U}_{NOT} = \hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (14)$$

Hadamard and Rotation gates

The Hadamard gate, which transforms the standard basis states into a balanced superposition

$$\hat{H} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (15)$$

Rotations of a qubit along the x, y, and z axes of the Bloch sphere:

$$\hat{R}_x(\theta) = e^{-i\theta \frac{X}{2}} = \begin{pmatrix} \cos(\frac{\theta}{2}) & -i\sin(\frac{\theta}{2}) \\ -i\sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{pmatrix} \quad (16)$$

$$\hat{R}_y(\theta) = e^{-i\theta \frac{Y}{2}} = \begin{pmatrix} \cos(\frac{\theta}{2}) & -\sin(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) & \cos(\frac{\theta}{2}) \end{pmatrix} \quad (17)$$

$$\hat{R}_z(\theta) = e^{-i\theta \frac{Z}{2}} = \begin{pmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{-i\frac{\theta}{2}} \end{pmatrix} \quad (18)$$

Any arbitrary single-qubit gate can be represented as a specific sequence of rotations around the z and y axes (Z-Y decomposition)

The *controlled-NOT* (*CNOT*) gate

Two-qubit gate where the target qubit remains unchanged if the control qubit is in the state $|0\rangle$; however, if the control qubit is in the state $|1\rangle$, the target qubit undergoes a bit-flip, as if a *NOT* gate had been applied

$$CNOT|00\rangle = |00\rangle$$

$$CNOT|01\rangle = |01\rangle$$

$$CNOT|10\rangle = |11\rangle$$

$$CNOT|11\rangle = |10\rangle$$

Its matrix representation is :

$$CNOT = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (19)$$

Quantum Circuits

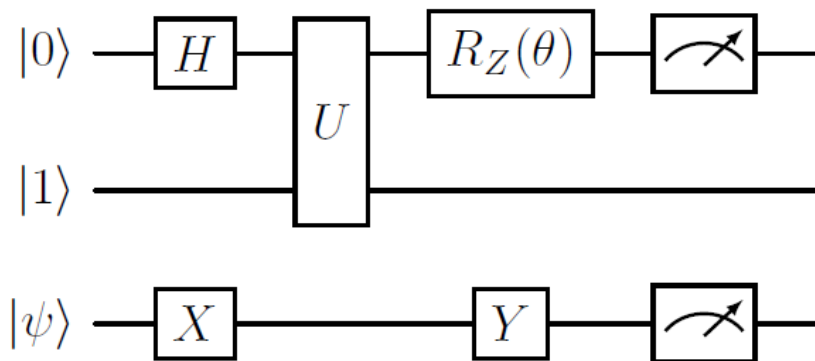


Figure: An example of a quantum circuit diagram

Input Encoding Techniques

Various techniques to **embed classical data** into an n -qubit quantum system:

- Basis** encoding

- Time-evolution** encoding

- Hamiltonian** encoding

- Amplitude** encoding

We will be using **Amplitude** encoding with **Amplitude-Efficient State Preparation** to encode 2^n features using n qubits

Variational Quantum Classifiers

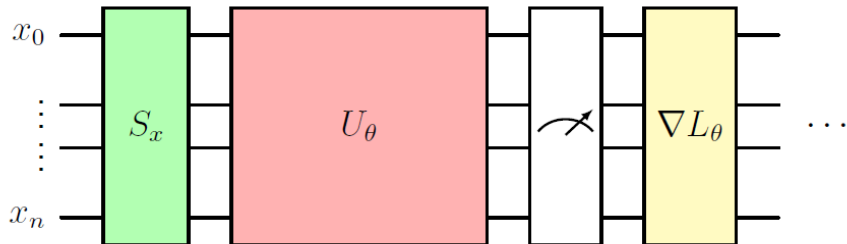


Figure: General structure of a variational quantum classifier

Breast Cancer

Most common malignancy among women worldwide

36% of all oncological cases

An estimated **2.089 million** women diagnosed in 2018

Lifestyle related factors:

Unhealthy dietary habits

Nicotine use

Chronic **stress**

Low levels of **physical** activity

Other risk factors:

Advanced **age**

Family **history** of breast cancer

Infection with **oncogenic** viruses

Early onset of **menstruation** or **late menopause**

First pregnancy at an **advanced** age or the **absence** of pregnancy

Use of **hormone replacement therapy**

Fine Needle Aspiration

We will be working with samples derived from Fine Needle Aspirates

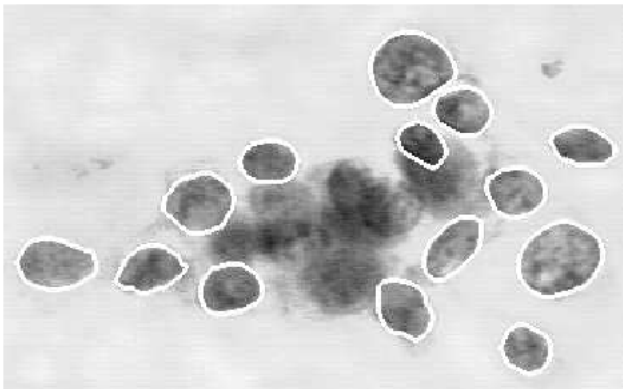


Figure: A magnified image of a malignant breast FNA with the visible cell nuclei having been outlined with the help of a curve fitting program.

Evaluation Metrics

Six models from before with **nine** key **metrics**:

$$\text{(Balanced) Accuracy } \left(\frac{1}{2} \left(\frac{TP}{TP+FN} + \frac{TN}{TN+FP} \right) \right)$$

$$\text{Precision } \left(\frac{TP}{TP+FP} \right)$$

$$\text{Recall } \left(\frac{TP}{TP+FN} \right)$$

$$\text{F1-Score } \left(2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2TP}{2TP+FP+FN} \right)$$

$$\text{MCC } \left(\frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \right)$$

$$\text{ROC AUC } \left(\int_0^1 \text{TPR}(\text{FPR}) d(\text{FPR}) \right)$$

$$\text{PR AUC } \left(\int_0^1 \text{Precision}(\text{Recall}) d(\text{Recall}) \right)$$

$$\text{Specificity } \left(\frac{TN}{TN+FP} \right)$$

$$\text{NPV } \left(\frac{TN}{TN+FN} \right)$$

Exploratory Data Analysis (feature distribution, class imbalance)

Finding the **best** model (80-20 split, rncv, data cleaning)

Fine tune the winner (bootstrapping, 95% confidence intervals)

Exploratory Data Analysis

30 features with **569** entries

Class **imbalance**: **357** instances of **benign**, **212** instances of **malignant** (62.74% - 37.25%)

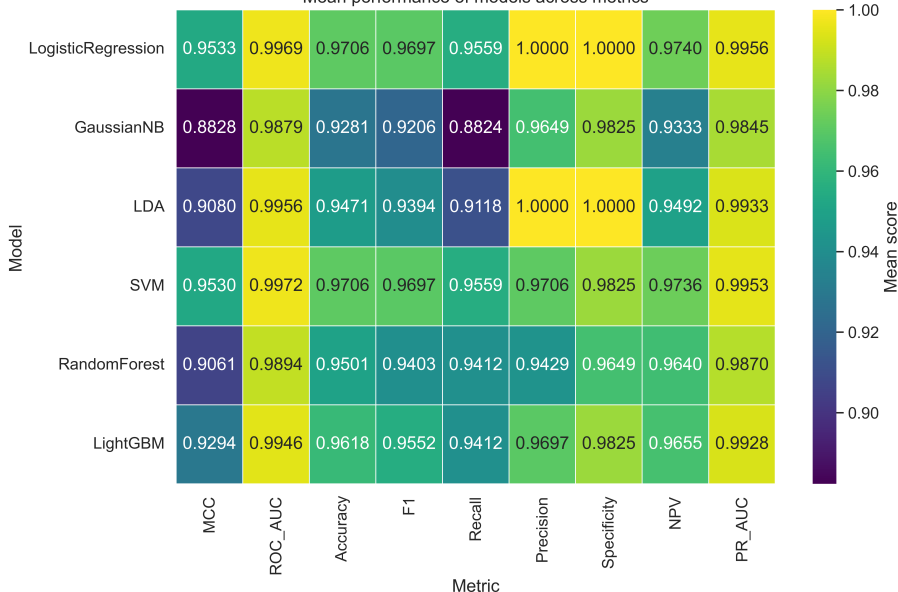
Majority of features exist on **different** scales, will **scale inside** rnCV

15 out of 30 features heavily **correlated** with **prognosis**

No correlation among features that we did not expect

Logistic Regression as the best model

Mean performance of models across metrics



Finetuning and bootstrapping

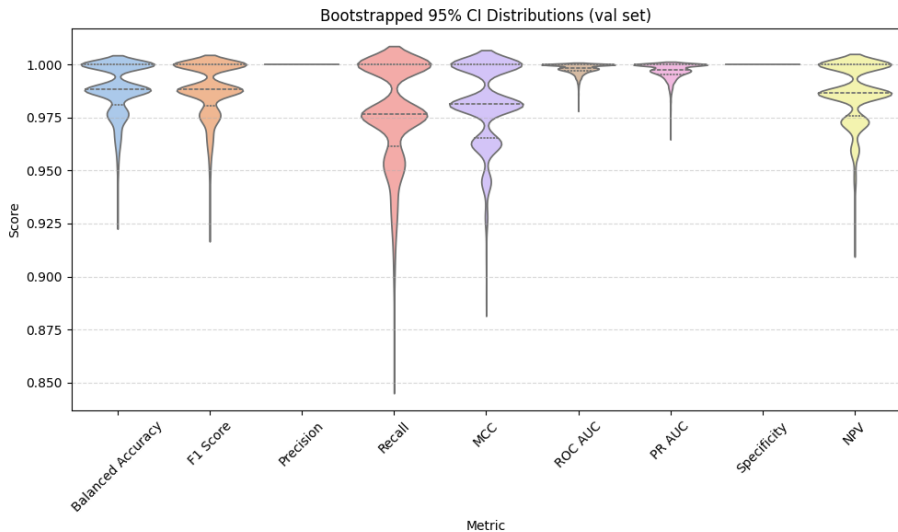


Figure: Violin plot of the confidence intervals

Quantum Experimental Framework

1, 2, 3, 4 and **5** qubits with **2, 4, 8, 16** and **30** features through **PCA**
60-20-20 train-validation-test split

-1 for **benign** and **1** for **malignant**

Mean Squared Error (**MSE**) for **loss** function, **Nesterov** Momentum for **optimizer**

Focus on **4** metrics: Accuracy, Recall, Precision, F1-score

Circuit consisting of **3** parts: Data **encoding**, **Variational** and **Measurement**

Same architecture for Variational: **Layers** consisting solely of **Rotations** and **CNOTs**

Two variants of VQCs: **Unbiased** vs **Biased**

Set number of training **epochs** for each **qubit**, we keep track of **time**
Average over **10** runs, keep **track** of the **best** performers per variation

One qubit results

Performance & Efficiency Trends: Biased vs Unbiased for 1 Qubits (10 Runs)

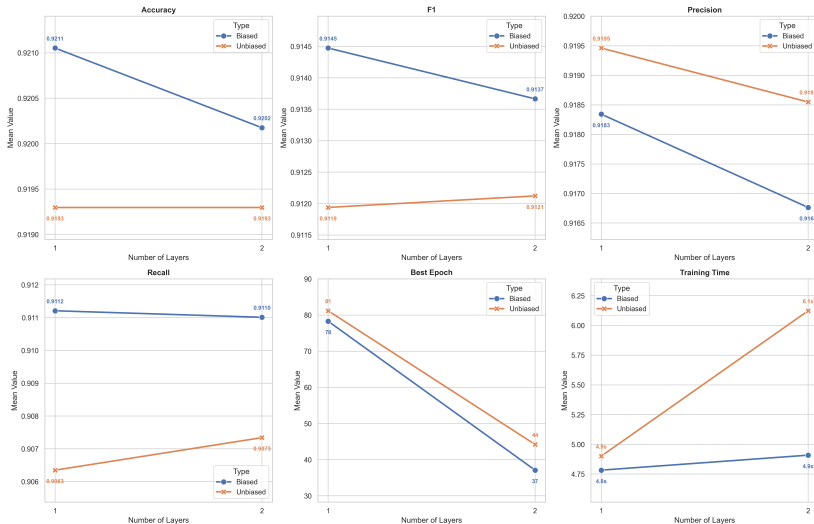


Figure: Performance and Efficiency Trends of Biased and Unbiased VQC for 1-Qubit

One qubit results, circuit of best unbiased VQC

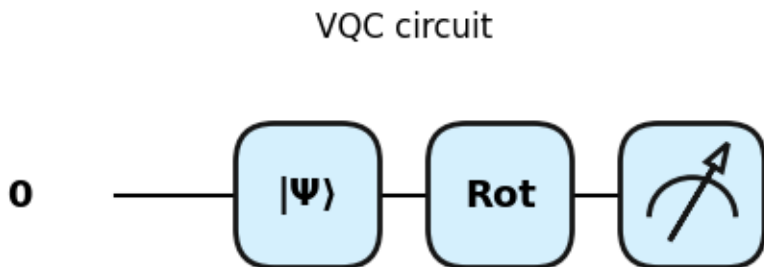


Figure: Circuit of the best unbiased 1-qubit VQC

One qubit results, loss evolution of best unbiased VQC

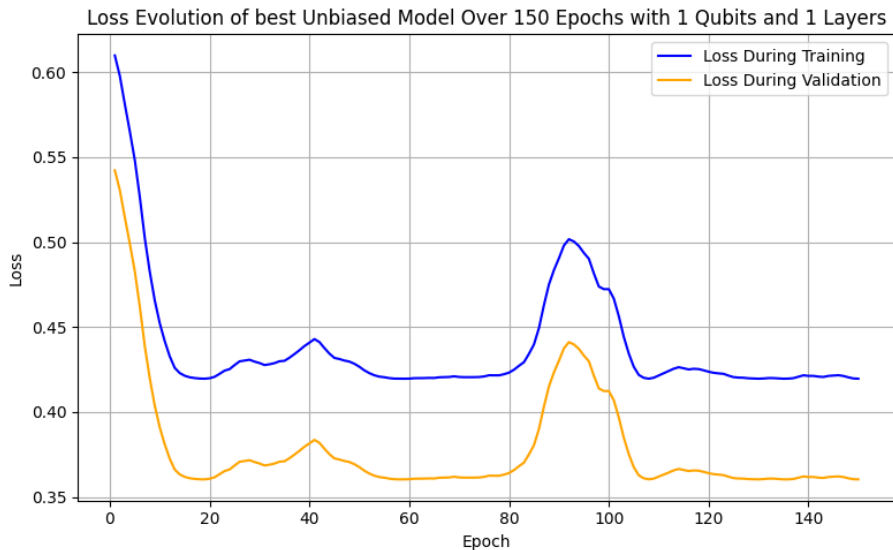


Figure: Evolution of the loss function of the best unbiased 1-qubit VQC

One qubit results, performance evolution of best unbiased VQC

Training vs Validation Metrics of best Unbiased model Over 150 Epochs for 1 Qubits and 1 Layers

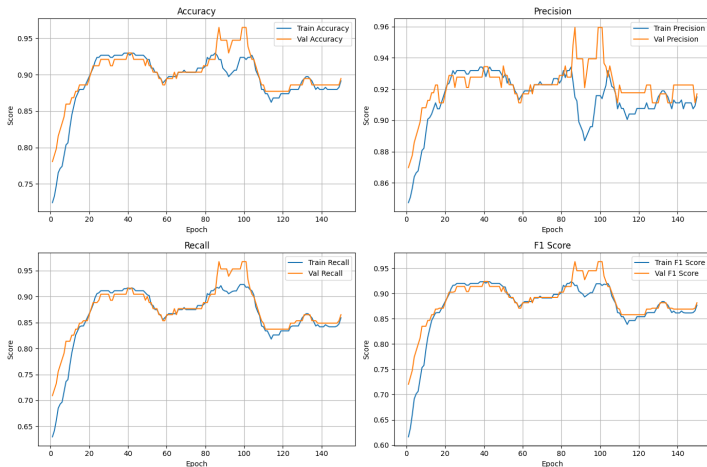


Figure: Validation and Training curves for the 4 key metrics of the best unbiased 1-qubit VQC

One qubit results, circuit of best biased VQC

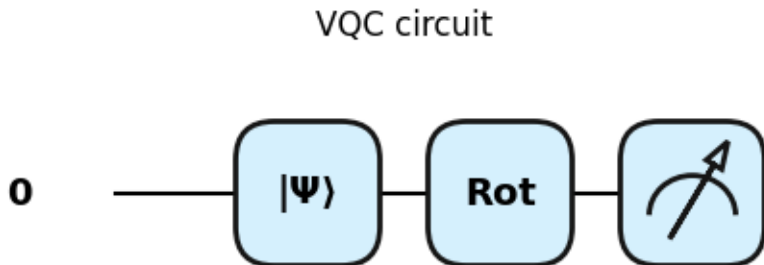


Figure: Circuit of the best biased 1-qubit VQC

One qubit results, loss evolution of best biased VQC

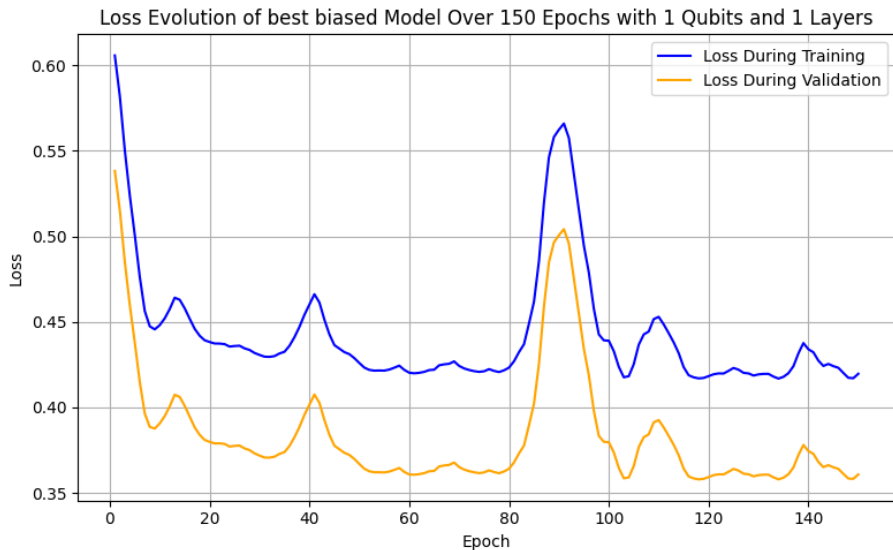


Figure: Evolution of the loss function of the best biased 1-qubit VQC

One qubit results, performance evolution of best biased VQC

Training vs Validation Metrics of best Biased model Over 150 Epochs for 1 Qubits and 1 Layers

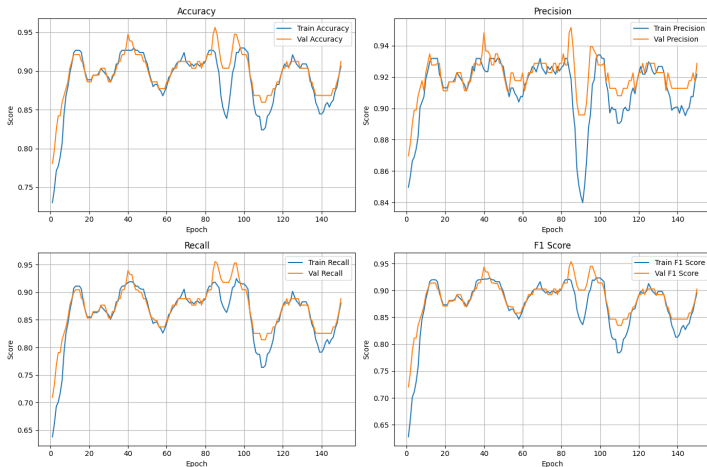


Figure: Validation and Training curves for the 4 key metrics of the best biased 1-qubit VQC

Two qubit results

Performance & Efficiency Trends: Biased vs Unbiased for 2 Qubits (10 Runs)

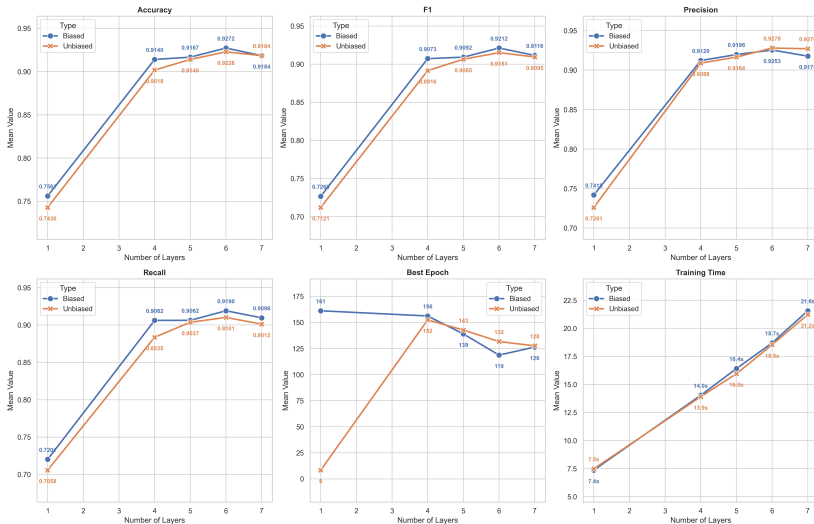


Figure: Performance and Efficiency Trends of Biased and Unbiased Classifier for 2 Qubits

Two qubit results, circuit of best unbiased VQC

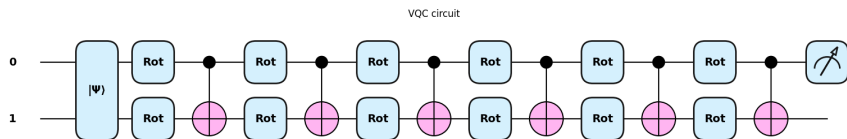


Figure: Circuit of the best 2-qubit unbiased classifier

Two qubit results, loss evolution of best unbiased VQC

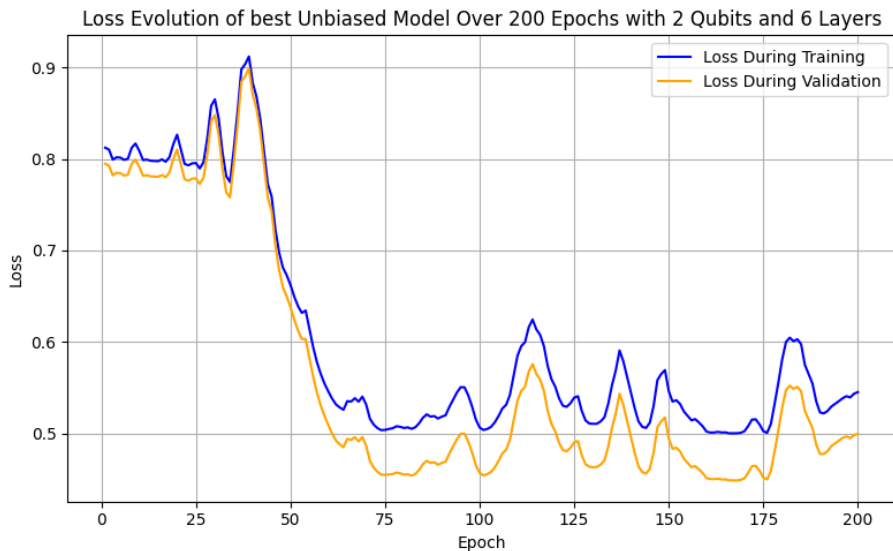


Figure: Evolution of the loss function of the best 2-qubit unbiased classifier

Two qubit results, performance evolution of best unbiased VQC

Training vs Validation Metrics of best Unbiased model Over 200 Epochs for 2 Qubits and 6 Layers

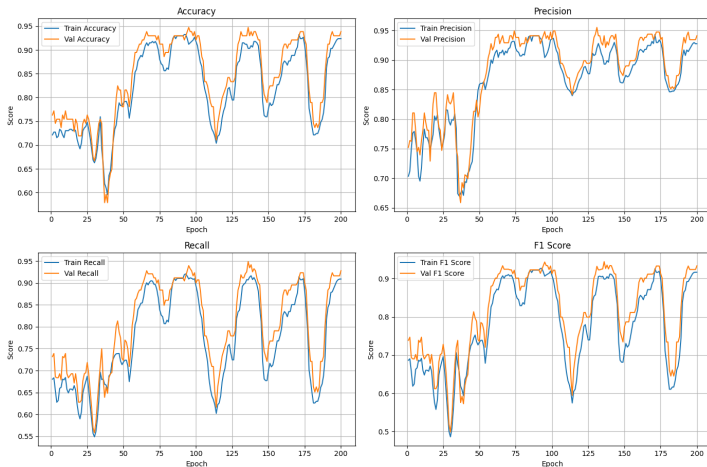


Figure: Validation and Training curves for the 4 key metrics of the best 2-qubit unbiased classifier

Two qubit results, circuit of best biased VQC

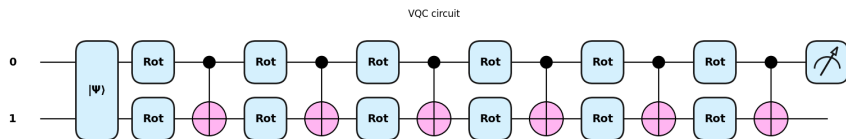


Figure: Circuit of the best 2-qubit biased classifier

Two qubit results, loss evolution of best biased VQC

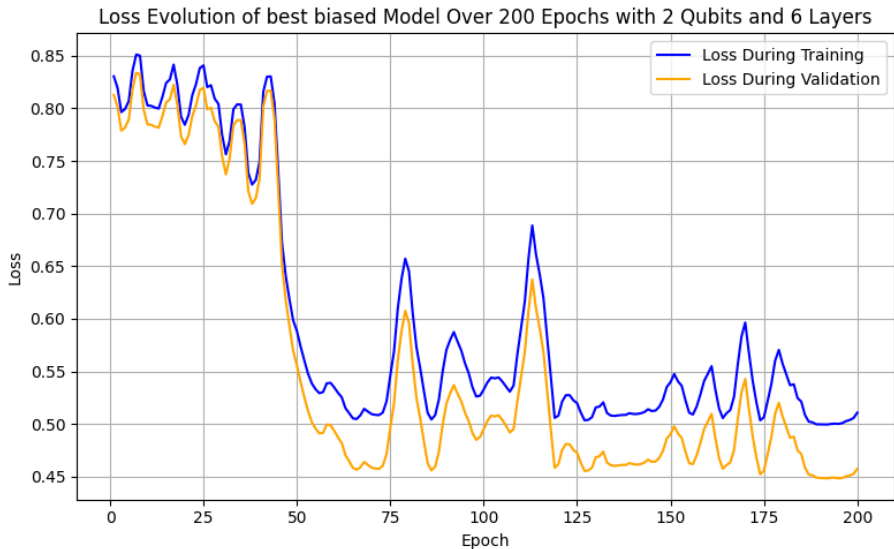


Figure: Evolution of the loss function of the best 2-qubit biased classifier

Two qubit results, performance evolution of best biased VQC

Training vs Validation Metrics of best Biased model Over 200 Epochs for 2 Qubits and 6 Layers

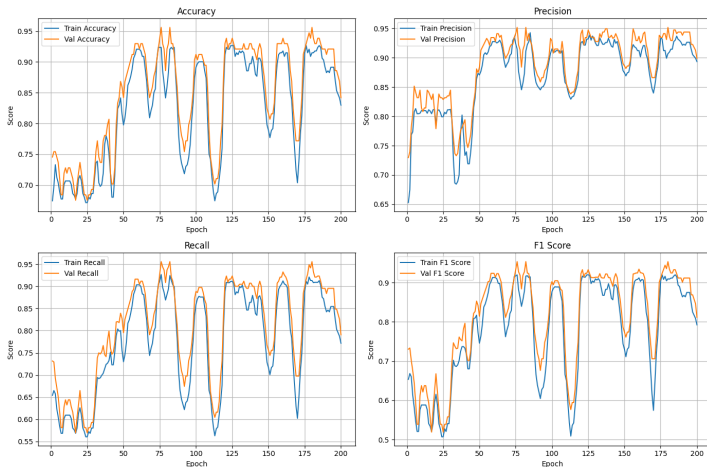


Figure: Validation and Training curves for the 4 key metrics of the best 2-qubit biased classifier

Three qubit results

Performance & Efficiency Trends: Biased vs Unbiased for 3 Qubits (10 Runs)

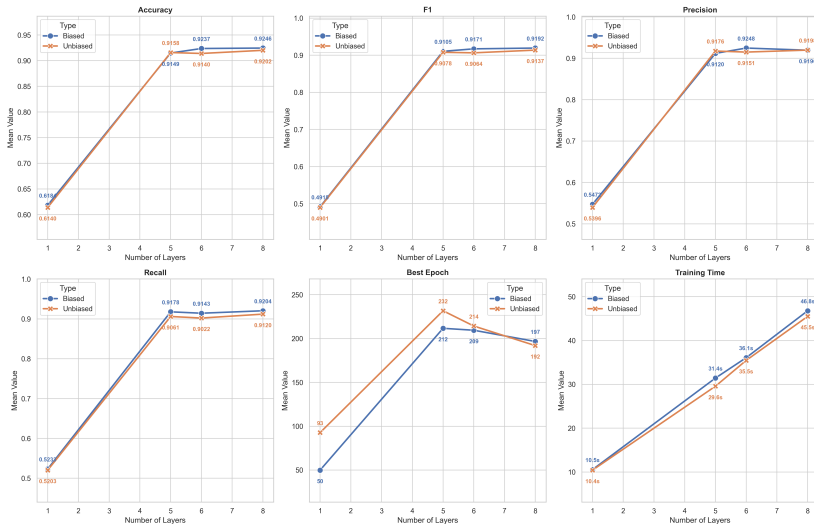


Figure: Performance and Efficiency Trends of Biased and Unbiased Classifier for 3 Qubits

Three qubit results, circuit of best unbiased VQC

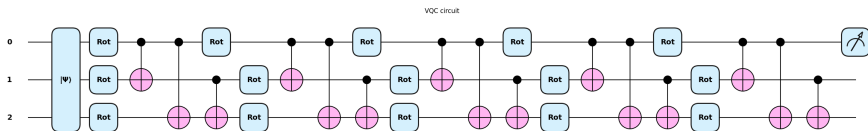


Figure: Circuit of the best 3-qubit unbiased classifier

Three qubit results, loss evolution of best unbiased VQC

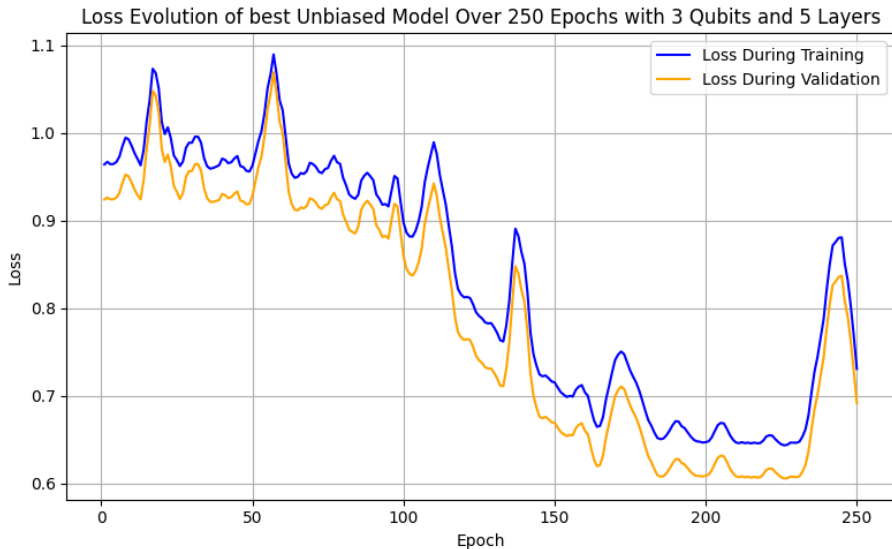


Figure: Evolution of the loss function of the best 3-qubit unbiased classifier

Three qubit results, performance evolution of best unbiased VQC

Training vs Validation Metrics of best Unbiased model Over 250 Epochs for 3 Qubits and 5 Layers

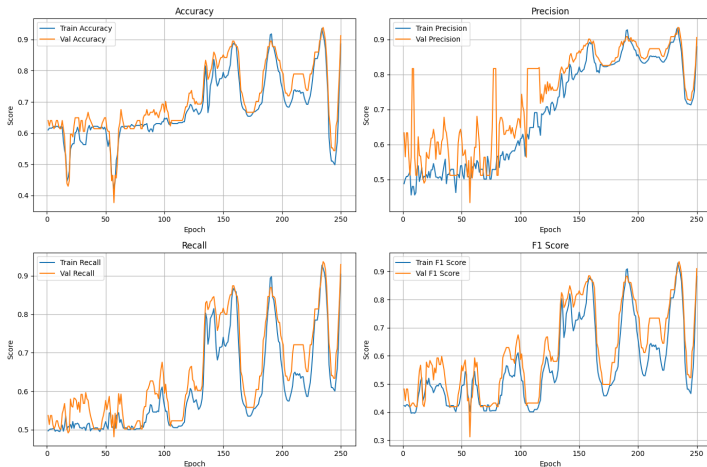


Figure: Validation and Training curves for the 4 key metrics of the best 3-qubit unbiased classifier

Three qubit results, circuit of best biased VQC

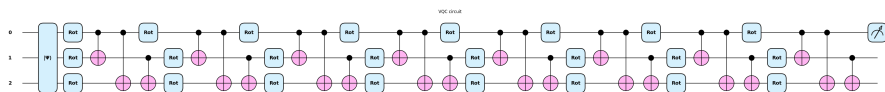


Figure: Circuit of the best 3-qubit biased classifier

Three qubit results, loss evolution of best biased VQC

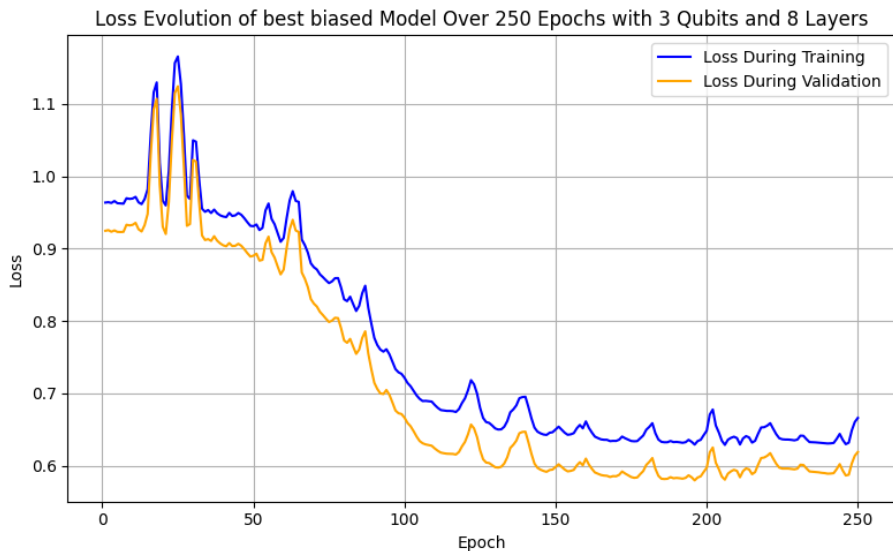


Figure: Evolution of the loss function of the best 3-qubit biased classifier

Three qubit results, performance evolution of best biased VQC

Training vs Validation Metrics of best Biased model Over 250 Epochs for 3 Qubits and 8 Layers

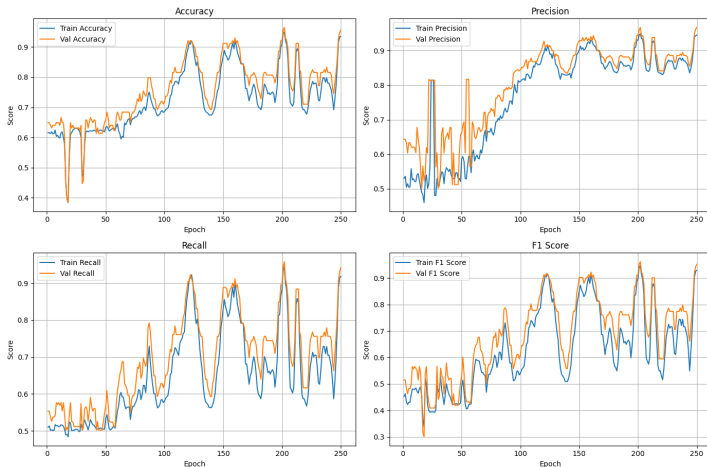


Figure: Validation and Training curves for the 4 key metrics of the best 3-qubit biased classifier

Four qubit results

Performance & Efficiency Trends: Biased vs Unbiased for 4 Qubits (10 Runs)

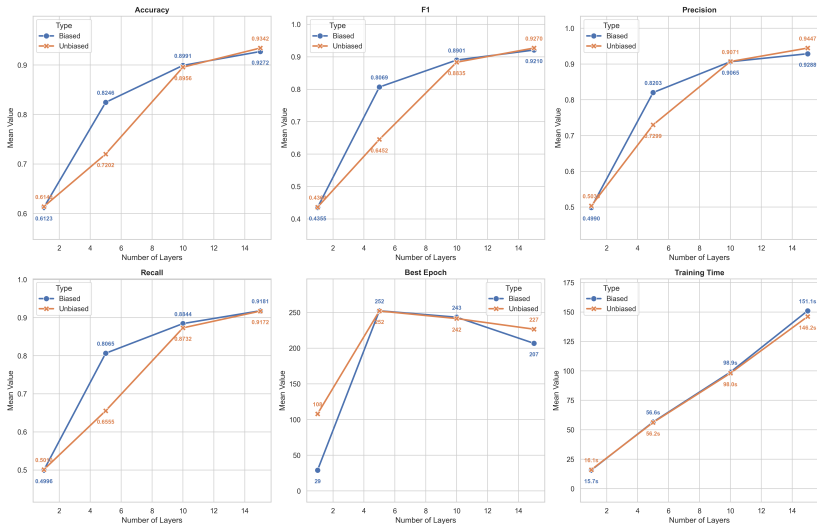


Figure: Performance and Efficiency Trends of Biased and Unbiased Classifier for 4 Qubits

Four qubit results, circuit of best unbiased VQC

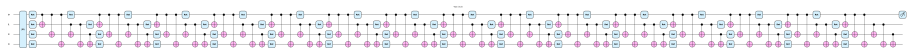


Figure: Circuit of the best 4-qubit unbiased classifier

Four qubit results, loss evolution of best unbiased VQC

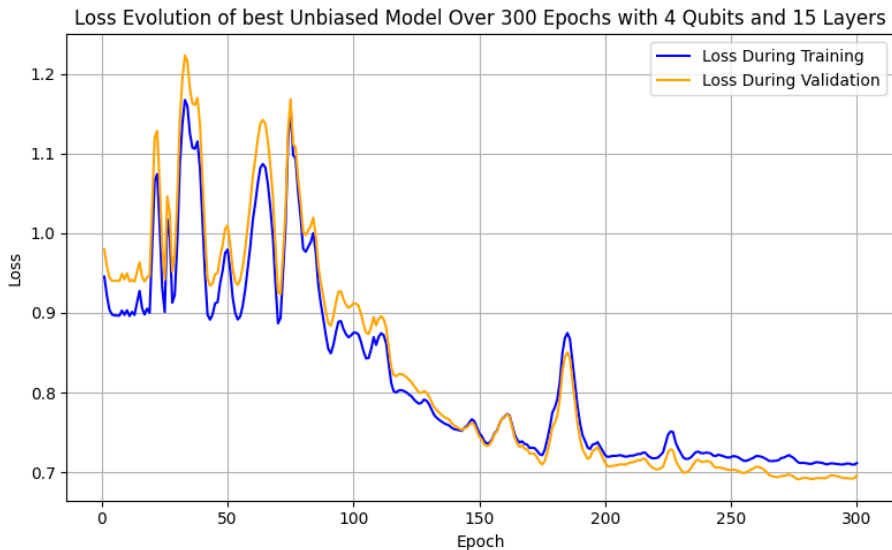


Figure: Evolution of the loss function of the best 4-qubit unbiased classifier

Four qubit results, performance evolution of best unbiased VQC

Training vs Validation Metrics of best Unbiased model Over 300 Epochs for 4 Qubits and 15 Layers

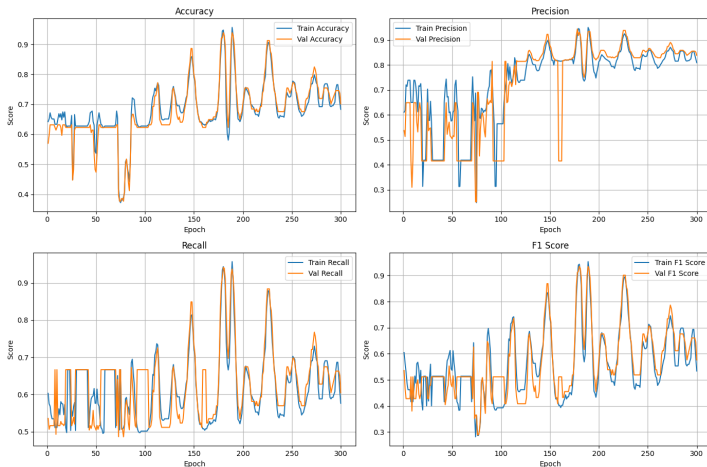


Figure: Validation and Training curves for the 4 key metrics of the best 4-qubit unbiased classifier

Four qubit results, circuit of best biased VQC

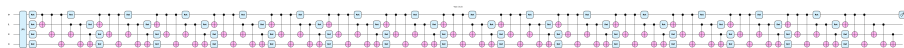


Figure: Circuit of the best 4-qubit biased classifier

Four qubit results, loss evolution of best biased VQC

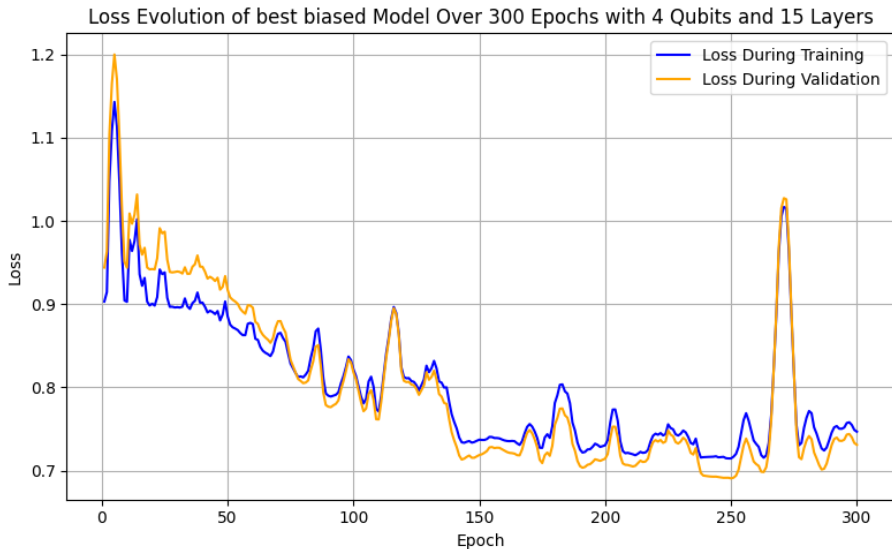


Figure: Evolution of the loss function of the best 4-qubit biased classifier

Four qubit results, performance evolution of best biased VQC

Training vs Validation Metrics of best Biased model Over 300 Epochs for 4 Qubits and 15 Layers

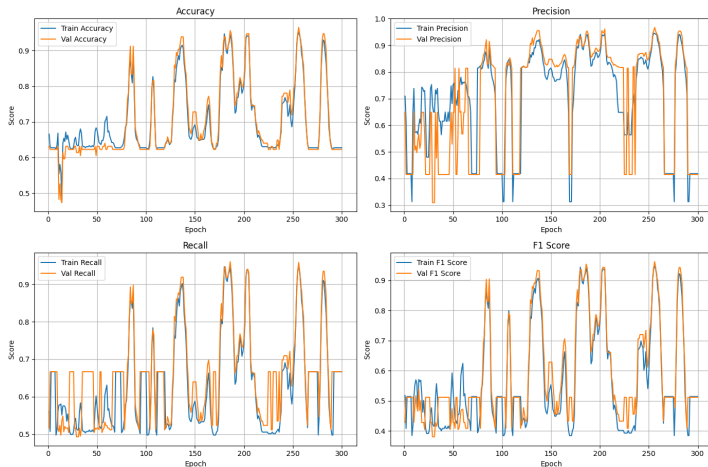


Figure: Validation and Training curves for the 4 key metrics of the best 4-qubit biased classifier

Five qubit results

Performance & Efficiency Trends: Biased vs Unbiased for 5 Qubits (10 Runs)

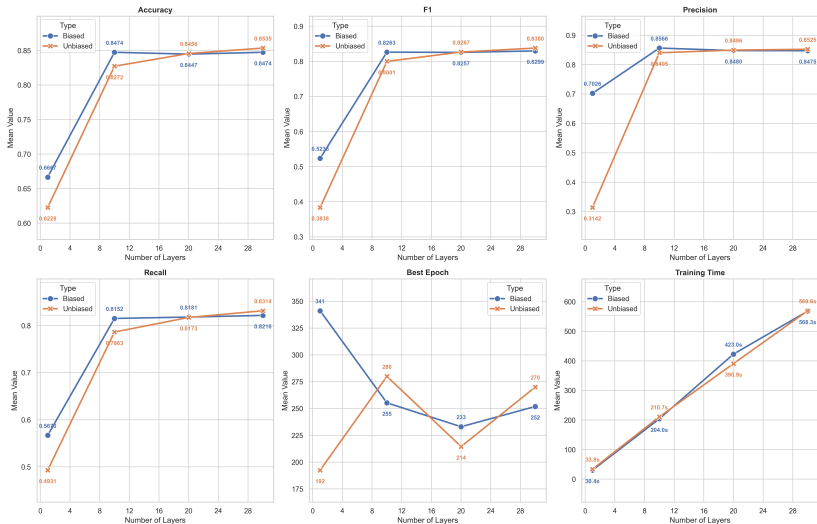


Figure: Performance and Efficiency Trends of Biased and Unbiased Classifier for 5 Qubits

Five qubit results, circuit of best unbiased VQC



Figure: Circuit of the best 5-qubit unbiased classifier

Five qubit results, loss evolution of best unbiased VQC

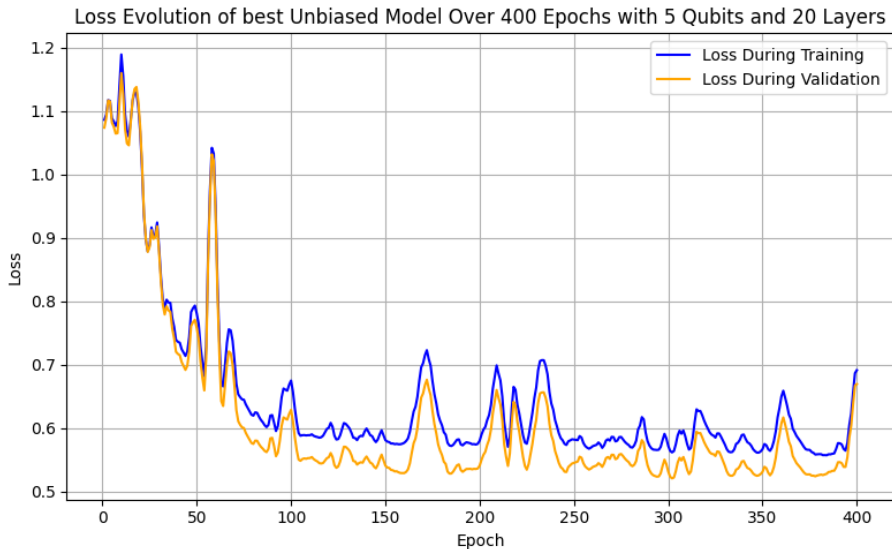


Figure: Evolution of the loss function of the best 5-qubit unbiased classifier

Five qubit results, performance evolution of best unbiased VQC

Training vs Validation Metrics of best Unbiased model Over 400 Epochs for 5 Qubits and 20 Layers

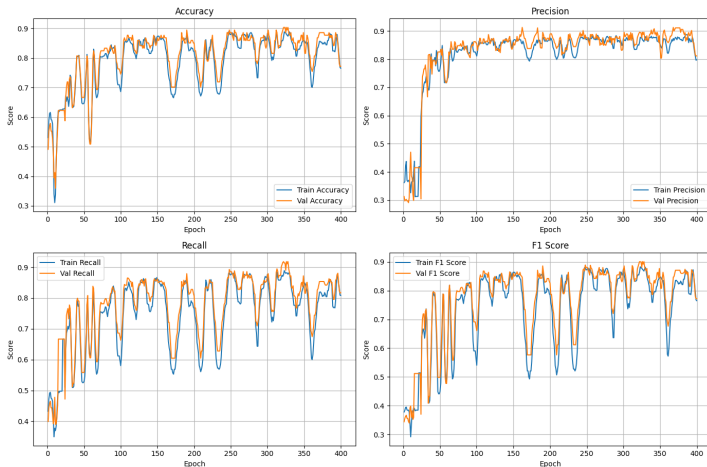


Figure: Validation and Training curves for the 4 key metrics of the best 5-qubit unbiased classifier

Five qubit results, circuit of best biased VQC



Figure: Circuit of the best 5-qubit biased classifier

Five qubit results, loss evolution of best biased VQC

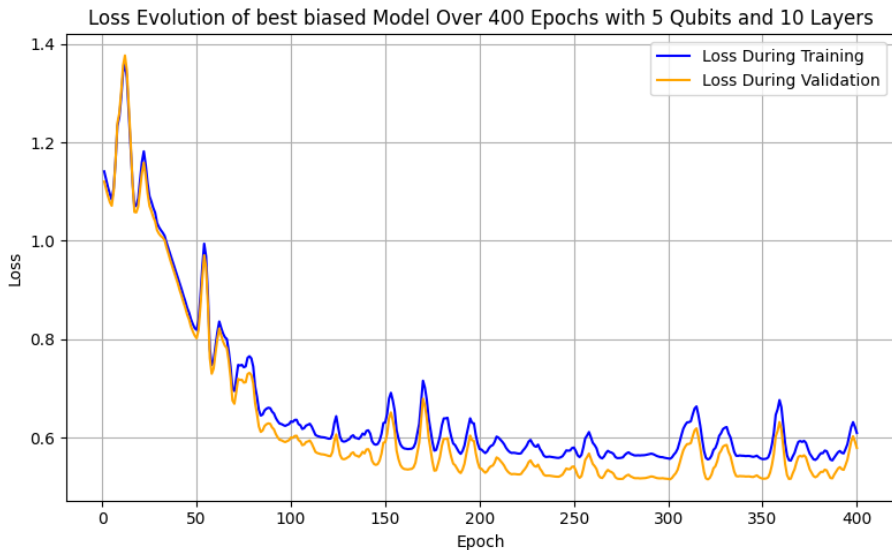


Figure: Evolution of the loss function of the best 5-qubit biased classifier

Five qubit results, performance evolution of best biased VQC

Training vs Validation Metrics of best Biased model Over 400 Epochs for 5 Qubits and 10 Layers

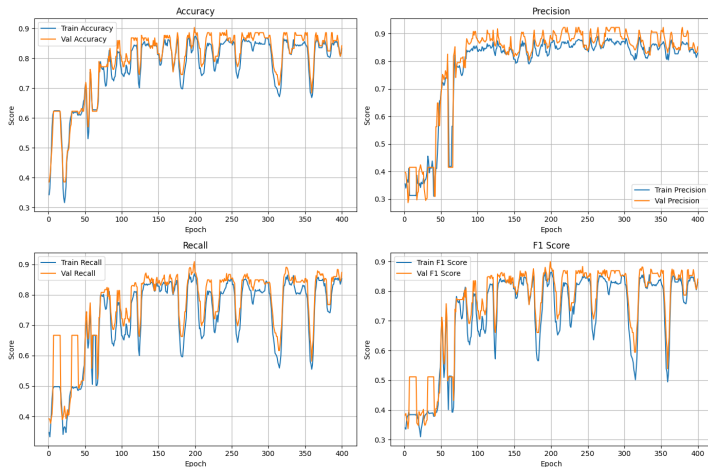


Figure: Validation and Training curves for the 4 key metrics of the best 5-qubit biased classifier

Comparison, Best unbiased performers

Unbiased VQC Metrics by Qubits (10 Runs)

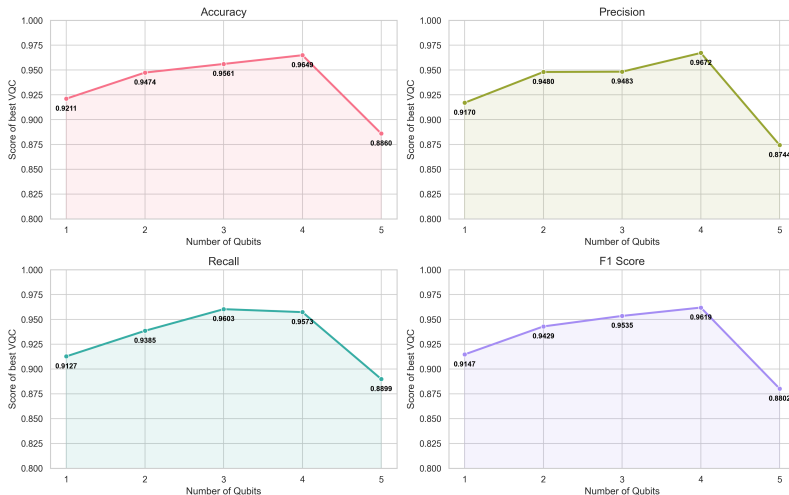


Figure: Performance of best unbiased classifier per number of qubits across all runs

Comparison, Best biased performers

Biased VQC Metrics by Qubits (10 Runs)

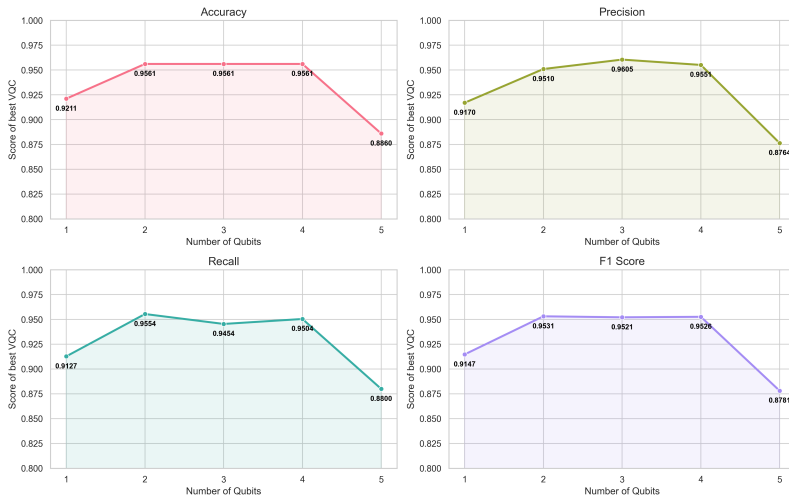


Figure: Performance of best biased classifier per number of qubits across all runs

Comparison, parameters needed for best performers

Parameter Efficiency: Unbiased vs Biased VQC (10 Runs)

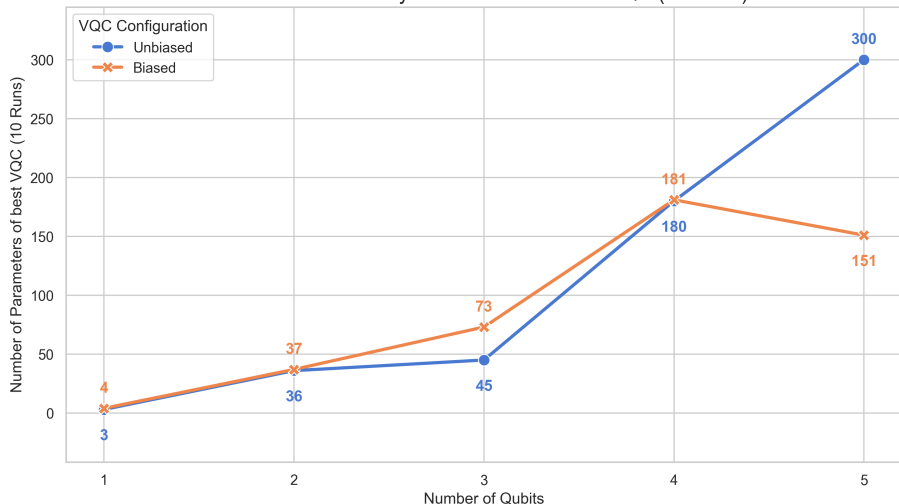


Figure: Number of trainable parameters in respect to the number of qubits for biased and unbiased classifiers

VQCs vs Logistic Regression

Average Performance Metrics: Classical ML vs VQCs

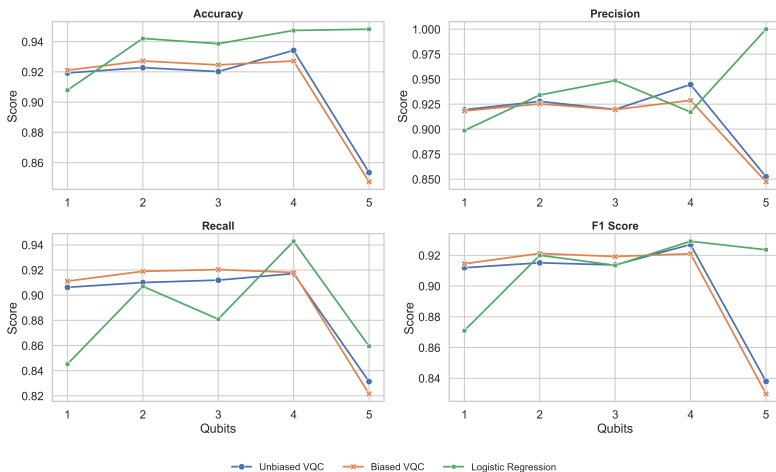


Figure: Average performance of the Quantum Models and the classical Logistic Regression model across 10 different runs for all the different number of qubits

VQCs show **strong potential**, especially in low dimensionality problems

Classical ML still **slightly** superior

Future quantum hardware will **improve** results

Thank you for your attention!

Questions?